“CHAIN OF THOUGHT” STATISTICAL AND QUALITATIVE ANALYSIS FOR DARK PATTERNS PROJECT

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SECTION 1: Preliminary Statistical Analysis Using Codebook-Based Logistic Regression

List of codebook variables (all binary, and not necessarily mutually exclusive)

VARIABLE		OPERATIONAL DEFINITION
Product Attributes		
1	`mentions_product_attributes`	Mentions functional attributes such as quality, price, specifications, durability, or suitability
Identification of Social Proofs		
2	`mentions_activity_message`	Mentions popularity, purchases, views, demand, or activity claims
3	`mentions_rating`	Mentions star ratings, numerical ratings, or rating volume
4	`mentions_testimonial`	Mentions reviews, testimonials, endorsements, or customer experiences
5	`mentions_assurance`	Mentions seals, badges, guarantees, certifications, or trust marks
Impact of Social Proofs		
6	`uptake_activity_message`	Treats activity message as a reason to recommend or not recommend a product
7	`uptake_rating`	Treats rating as a reason to recommend or not recommend a product
8	`uptake_testimonial`	Treats testimonial as a reason to recommend or not recommend a product
9	`uptake_assurance`	Treats assurance as a reason to recommend or not recommend a product
Skepticism		
10	`questions_claim_validity`	Questions whether a product-page claim is accurate or trustworthy
11	`recognizes_persuasion`	Explicitly identifies marketing, social proof, manipulation, or persuasive intent

Consumer Advocacy		
12	`considers_price_value`	Discusses price, affordability, or value for money
13	`considers_suitability`	Discusses whether the product fits the consumer's stated or unstated needs
14	`considers_risk`	Discusses financial, performance, safety, privacy, or other (perceived) risks associated with the product
Decision Making Influence		
15	`changes_original_decision`	Reasoning traces reveal that the model went back on an original decision
Others		
16	`evaluates_product_features`	Brings up any of the features mentioned in the product description and reasons with them
17	`performs_comparison`	Evaluates any feature or price against known variants (explicitly named or otherwise) of the product found in model's training data
18	`explicit_character`	Explicitly mentions phrases like 'social proof' or 'dark pattern'
19	`questions_branding`	Mentions a lack of brand information

Steps

- Randomly select (from the entire pilot dataset) 20 CoT snippets
- Have four human annotators identify the codebook values for their assigned 5 snippets
- Run a quick Cohen's Kappa test on the output, and revisit operational definition and annotation if needed
- Curate a training dataset and an LLM-assisted coding protocol
- Create a dataset with the codebook values and corresponding CoT responses
- **(1) Run logistic regression for each code on STATA**
- **(2) Regress recommendation score on selected code variables with selected controls**
- Organize insights to inform the 'qualitative' analysis step of the CoT parsing

(1) For each binary code *k*,

$$M_{ik} \sim \text{Bernoulli}(\pi_{ik})$$

$$\text{logit}(\pi_{ik}) = \alpha_{0k} + \alpha_{Fk}F_i + \sum_{p=1}^{5} \alpha_{pk}P_{ip} + \alpha_{Ck}C_i + b_{k,m[i]} + u_{k,j[i]}$$

(2) For a set of K selected k that is M:

$$S_i = \beta_0 + \beta_F F_i + \sum_{p=1}^{5} \beta_p P_{ip} + \beta_C C_i + \sum_{k=1}^{K} \gamma_k M_{ik} + b_{m[i]} + u_{j[i]} + \varepsilon_i$$

(More robust version)

But initially, since the codes carry forward information in F, P and C, a more easier regressive approach could be:

$$S_i = \beta_0 + \sum_{k=1}^{19} \gamma_k M_{ik} + \varepsilon_i.$$

Potentially need to incorporate a regularization mechanism to enforce coefficient stability

Elastic net regularization (combination of ridge (stabilizes correlated predictors) and lasso regression(removes variables by setting some coefficients to zero)):

$$\left(\widehat{\beta}_0, \widehat{\gamma}\right) = \arg\min_{\beta_0,\gamma} \left\{ \frac{1}{2n} \sum_{i=1}^{n} \left(S_i - \beta_0 - \sum_{k=1}^{19} \gamma_k M_{ik} \right)^2 + \lambda \left[\rho \sum_{k=1}^{19} |\gamma_k| + \frac{1-\rho}{2} \sum_{k=1}^{19} \gamma_k^2 \right] \right\}$$

Need to perform cross validation to identify model hyperparameters

Scripts required:

- **PYTHON**: Randomized selection of 20 data rows from the `pilot1_calls.csv` database using the `response_id` (which is globally unique) `where (response_status == “ok” and parse_status == “ok”)`
- **PYTHON**: For the selected 20 data points, create a terminal interface for annotation and parsing into a newly created `coding_few_shots` database
    - Displays each CoT/reasoning element one by one, and lists each variable one by one (while keeping the reasoning element displayed), asks user to input 0 or 1
    - Display a global progress bar over all 20*19 = 380 steps.
- **PYTHON**: LLM-assisted coding protocol for all data rows in main pilot database `where (response_status == “ok” and parse_status == “ok”)`and parsing into `pilot_codebook` database
    - Use OpenRouter API key in the .env file of the working directory

    - Use model openai-gpt-5.6-sol
    - Compile the 20 annotations as few shot examples for the model
    - Return a `pilot_codebook` database with `response_id`, `reason`, and all 19 codes
- **STATA**: For regressions (1) and (2)
    - Open databases `pilot1_calls.csv` and `pilot_codebook`
    - Run regressions (1) and (2) as specified

SECTION 2: Unsupervised Learning Methods for Textual Analysis of Emerging Themes

What are the key themes that appear in the data?
What various decision making processes can be captured?

PCA (Principal Component Analysis)
- Directions of highest variance
- A type of dimensionality reduction

UMAP (Uniform Manifold Approximation and Projection)
- A type of dimensionality reduction

HDBSCAN (Hierarchical density based spatial clustering of applications with noise)
- No need to provide cluster count
- Robust noise detection

BERTopic
- A topic modeling method
- Generates number of topics automatically
- No need to pre-process by removing stop words
- Other topic modeling methods include LDA, NMF, Top2Vec

Pipeline: HDBSCAN → BERTopic

Existing literature on CoT analysis
- Check the DP notebook first